

Data Integration

Four datasets were successfully merged into a master dataset which included features like area(country), year, average_rain_fall_mm_per_year, avg_temp, value_x(pesticide use), value_y(crop yield).

Exploratory Insights

From the EDA:

- the master dataset spans multiple countries (about 137),
- rainfall and temperature varied by country and year,
- average rainfall ranges between 300mm/year to 2000mm/year, while temperature ranges between 10°C and 30°C,
- pesticide use and yield differ among countries.

Relationships

Based on correlation analysis:

- Moderate rainfall generally increases yield
- High temperatures reduce yield
- Moderate pesticide use lead to better yield
- Countries with more stable climates show better yield

Model

A regression model was built and trained. It was realized that extreme temperature reduces yield, more pesticides application resulted to a higher yield (indicating pest control efficiency), while sufficient rainfall (the strongest predictor) improves yield.

Implications for farmers

- The farmers should optimize the pesticide application to reach target yield levels without waste.
- If predicted yield crops drops, the farmers should plan for alternate crops or protective measures.
- The farmers should identify regions where extreme temperature or rainfall stress reduces yield and shift planting calendars.

Conclusion

The model demonstrates that rainfall and temperature are the most important environmental factors while the pesticide use was a key agricultural factor. The final predictive model gives farmers actionable yield forecasts to help them to guide resource use, reduce risk and improve yield and/or profit.